

Homework on data types

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Fill in the Blanks

1. The bool data type in Python contains the values _____ and _____.
2. A list in Python is enclosed within _____ brackets.
3. Tuple elements are enclosed within _____.
4. Dictionary data is stored in the form of _____ pairs.
5. The operator // is known as _____ division operator.
6. The input() function accepts data as a _____ by default.
7. The ** operator is used for _____ operation.
8. Variables whose values cannot be changed are called _____ variables.
9. The operator used to check membership in a sequence is _____.
10. The process of removing errors from a program is called _____.

True / False

1. Python sets allow duplicate values.
2. Strings in Python can be enclosed in single or double quotes.
3. Tuples are mutable data types.
4. The not operator is a logical operator.
5. Dictionary keys are separated from values using : symbol.
6. The input() function automatically converts numeric input into integers.
7. The is operator checks whether two variables refer to the same object.
8. Lists are ordered collections in Python.
9. None represents the absence of a value.
10. Runtime errors occur during program execution.

Multiple Choice Questions (MCQs)

1. Which of the following is a mutable data type?
 - a) Tuple
 - b) String
 - c) List
 - d) Integer
2. Which operator is used for exponentiation in Python?
 - a) ^
 - b) **
 - c) //
 - d) %%
3. Which data type stores key-value pairs?
 - a) Set
 - b) List
 - c) Dictionary
 - d) Tuple
4. Which operator checks equality between two values?
 - a) =
 - b) ==
 - c) !=
 - d) >=

5. Which function converts a value into integer type?
 - a) str()
 - b) float()
 - c) int()
 - d) bool()
6. Which operator is used for floor division?
 - a) /
 - b) %
 - c) //
 - d) **
7. Which of the following is an immutable sequence?
 - a) List
 - b) Set
 - c) Dictionary
 - d) Tuple
8. Which operator returns remainder after division?
 - a) //
 - b) %
 - c) /
 - d) **
9. Which logical operator returns True if both operands are True?
 - a) or
 - b) and
 - c) not
 - d) is
10. Which function is used to display output in Python?
 - a) display()
 - b) printf()
 - c) print()
 - d) output()

Short Questions

1. Define mutable and immutable data types.
2. What is the difference between list and tuple?
3. Explain the purpose of the input() function.
4. What are arithmetic operators in Python?
5. Define dictionary with an example.
6. What is type conversion in Python?
7. Explain identity operators with examples.
8. What is operator precedence?
9. Define debugging in Python programming.
10. Differentiate between syntax error and runtime error.

Coding Questions

1. Write a Python program to demonstrate all arithmetic operators.
2. Develop a program to check whether a given element exists in a list using membership operators.
3. Write a Python program to create a dictionary containing student details and display them.

4. Create a program to perform explicit type conversion on integer, float, and string values.
5. Write a program to demonstrate the use of logical operators using user input.
6. Develop a Python program to find the largest among three numbers using relational operators.
7. Write a program to demonstrate mutable and immutable data types.
8. Create a Python program to take user input for name and age and display formatted output using print().
9. Write a program to perform various assignment operations (+, -, *, /) on a variable.
10. Develop a Python program to demonstrate operator precedence using different expressions.